

Design of a Modular Digital Data Monitoring Unit

Phase-I RTL, Simulation and Xilinx Vivado Synthesis Report

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Abstract—The embedded systems that require real-time digital monitoring can overlook real-time abnormal occurrences if software polling is used. The Phase-I framework describes an 8-bit Digital Event Monitoring Unit (DEMU) with features of programmable threshold comparison, signed/unsigned modes of data, sticky alarm functionality, fault value recording and a three-state finite-state machine (FSM). Original Phase-I report is related to the usable monitoring IP and its areas of usage. This report presents the Phase-I RTL results obtained from AMD/Xilinx Vivado 2024.1 by means of the used behavioral simulation, RTL composition, technology-mapped synthesis, resource use, and timing analysis. Spartan-7 XC7S25CSGA225-1 was chosen as the synthesis target. The design meets a user timing constraint of 10 ns (i.e. 100 MHz) and achieves WNS of +6.720 ns and WHS of +0.159 ns, with no failures. The design synthesis uses 26 slice LUTs, 12 slice registers, 30 bonded I/O blocks, and 1 BUFGCTRL item.

Index Terms—Digital Event Monitoring, Verilog HDL, Sticky Alarm, Fault Capture, FSM, RTL, FPGA Synthesis, Spartan-7, Vivado.

I. INTRODUCTION

The rapid changes in IoT and Industry 4.0 systems lead to the threat of sensor data being lost in the Phase-I development. The first project proposes re-use of the Intellectual Property (IP) core for the role of the hardware watchdog for the sensors' data. In fact, the initial motivation for this project observes the drawback of the processor-based monitoring: when the incident occurred, the processor was probably busy, that is why the traditional monitoring logic can receive the information on the incident and store it for later diagnosis by the software.

The Phase-I project considers three issues: loss of transient events, lack of efficiency of traditional alarms, and static configuration of the monitoring device to monitor one type of data only. The stated features are latching alarm, value capturing, and choice of the dynamically changing data type.

The report presents the objectives for Phase-I design and describes the extra validation in terms of Xilinx Vivado conducted for the Phase-I RTL.

II. PHASE-I DESIGN OBJECTIVES

The main goal is to design a modular 8-bit monitoring device that can constantly compare sensor information with a user-set limit and deal with unusual situations through an FSM. The design provides:

- Signal from the 8-bit sensor and threshold.
- Signed and unsigned monitoring.
- Three-state FSM including IDLE, ALARM, and COOLDOWN.
- Active alarm until acknowledged by software.

- Capture of faults for the faults at the sensor value causing the alarm.
- Synchronous operation with an asynchronous active-low reset.

The original architecture phase one spells out seven functional blocks, which are: Sensor Register, Threshold Register, MUX Selector, Comparator, FSM Controller, Alarm Register, and Fault Capture Register. These blocks carry out the monitoring task by way of performing input processing tasks, comparison, control state, and diagnostics storage.

III. SYSTEM SPECIFICATIONS

Table I The specifications outlined in the original report are summarized in this document.

TABLE I: Phase-I system specifications

Parameter	Value / Description
Clock frequency	50 MHz stated as the average maximum processing speed
Sample latency	1 clock cycle for producing an alarm after detection
FSM states	IDLE, ALARM, COOLDOWN
State registers	2 bits
Reset	Asynchronous, active-low
Data width	8 bits for sensor and threshold data
Interface	Operations synchronized with clock

For the Vivado timing experiment mentioned here, a clock constraint of 10 ns was used (100 MHz target period). This is a timing constraint. It should not be deemed as a replacement for the 50 MHz operating speed mentioned in the original specifications from Phase I.

IV. RTL IMPLEMENTATION

The implementation of Phase-I RTL signifies Verilog HDL usage. The main module showcases the clock, asynchronous reset, threshold, monitor enable, data mode, sensor data, software acknowledgement, alarm output and fault capture output.

The FSM uses the following encodings:

```
idle_state = 2'b00;  
alarm_state = 2'b01;  
cooldown_state = 2'b10;
```

IDLE concept has active monitoring, which helps in knowing whether the comparison value will trigger the entry into ALARM. When in ALARM, the device keeps the alarm on until there is an acknowledgment by the software. The state machine will then proceed into COOLDOWN where the value

being monitored is expected to move to the acceptable side of the limit for the system to get back into IDLE.

The RTL design also has separate paths for signed and unsigned conversion. The fault capturing register is also used to capture the sensor reading during ALARM to provide diagnostic information regarding the event that has triggered the alarm.

V. RTL ELABORATION IN VIVADO

Phase-I RTL was brought to AMD/Xilinx Vivado 2024.1 and it was elaborated using Spartan-7 XC7S25CSGA225-1 as the target device. The elaborated design comes with comparison logic, FSM next-state multiplexing, state register, alarm register, and fault capture register.

Since the full-fledged generated schematic is very long and dense, it is divided into 3 sections below. These three figures are the crops of the same output of the elaborated design from Vivado which should be read together.

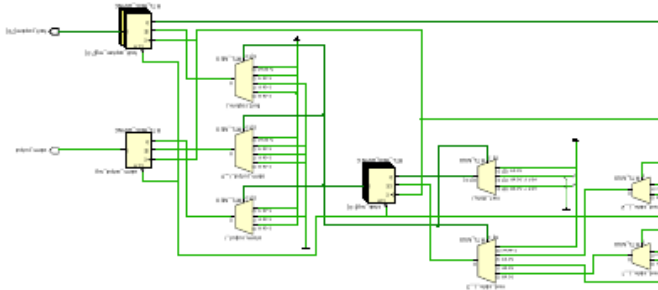


Fig. 1: Elaborated RTL schematic – upper section.

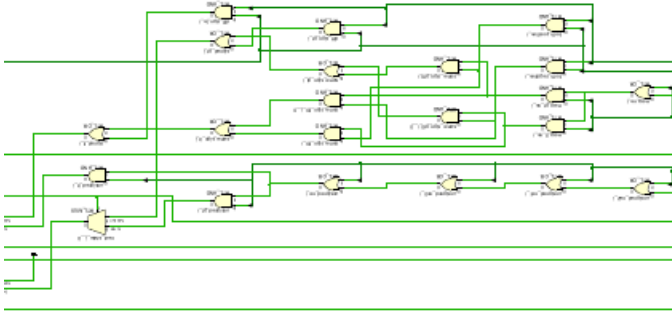


Fig. 2: Elaborated RTL schematic – middle section.

The detailed diagram is inclusive of RTL subtraction as well as ANDOR and multiplexer models for comparison and state selection logic. The FSM is modeled using a bit state register with next state multiplexing as well as registering of the alarm and fault-capturing outputs.

VI. BEHAVIORAL SIMULATION

In the given Phase-I testbench, a 10 ns clock is generated via a 5 ns half-period, and it engages with 8 application-specific verification tasks. The scenarios involved include industrial motor overheating, automotive batteries being overcharged,

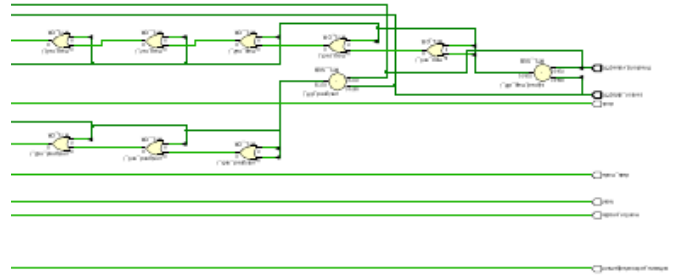


Fig. 3: Elaborated RTL schematic – lower section.

issues with smart thermostats, robotics collage and current surges, medical monitoring issues, environmental flood warnings, audio clipping, and issues related to security or intrusion.

During the simulation process, the waveform illustrated in Fig. 4 shows the variations in the threshold and readings of the sensor, wherein monitor enablement, data mode, software acknowledgment, alarm output, and fault capturing. The alarm is enabled when there is a fault, hence retaining the faulty output which is part of the sticky monitoring process.

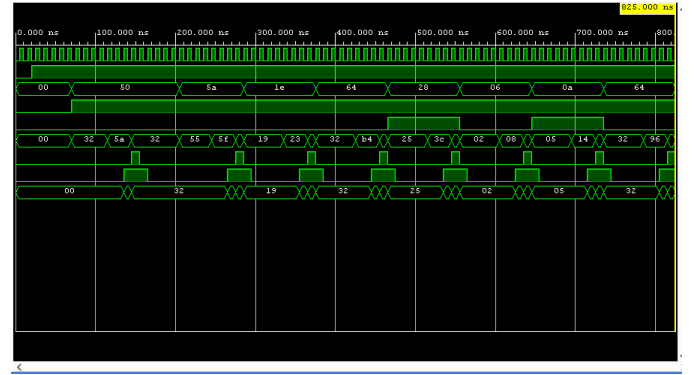


Fig. 4: Vivado behavioral simulation waveform for the Phase-I data monitor.

The testbench explicitly checks the captured fault value after each triggering event. This provides a direct functional check of the fault-capture mechanism across the eight application scenarios.

VII. XILINX VIVADO SYNTHESIS

A. Synthesis Target and Flow

The verification methods of the testbench confirm explicitly that captured fault value is verified in each occurrence of the test signal event. Thus, we verify functioning of fault-capture functionality during eight particular applications.

Synthesis uses AMD/Xilinx Vivado version 2024.1 and Spartan-7XC7S25CSGA225-1 target device. The process has not involved any physical expansion board use. Therefore, there is no need to perform any board-specific generation of pin assignments and/or bitstreams. Static timing analysis must be done with the use of the XDC constraint which provides the clock period.

The synthesis procedure is as follows:

- 1) Add the Phase-I RTL as a part of the design source.
- 2) Add the Phase-I testbench as a part of the source for simulation.
- 3) Analyze the RTL and check the RTL schematic.
- 4) Carry out the synthesis for Spartan-7 target.
- 5) Check the technology-mapped schematic.
- 6) Create reports on utilization and timing.

B. Synthesized Schematic

Subsequent to synthesis, Vivado associated the RTL to Spartan-7 resources like lookup tables (LUTs), flip-flops, CARRY4 units, I/O buffers and a global clock buffer. Similar to RTL hierarchy the synthesized design has been divided into several parts that are conducive to report writing.

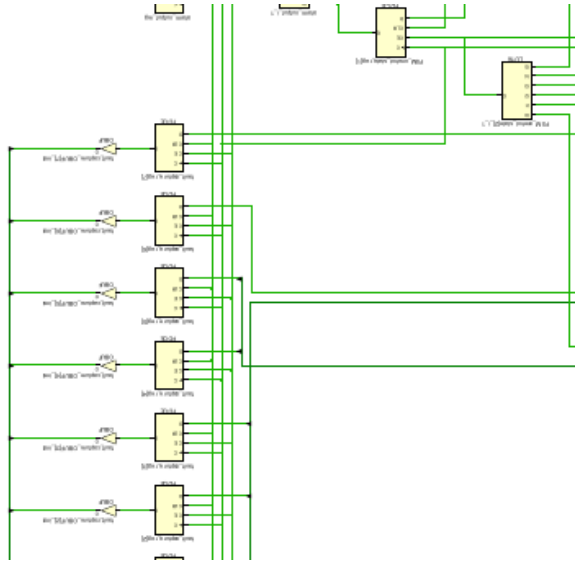


Fig. 5: Synthesized schematic – upper section.

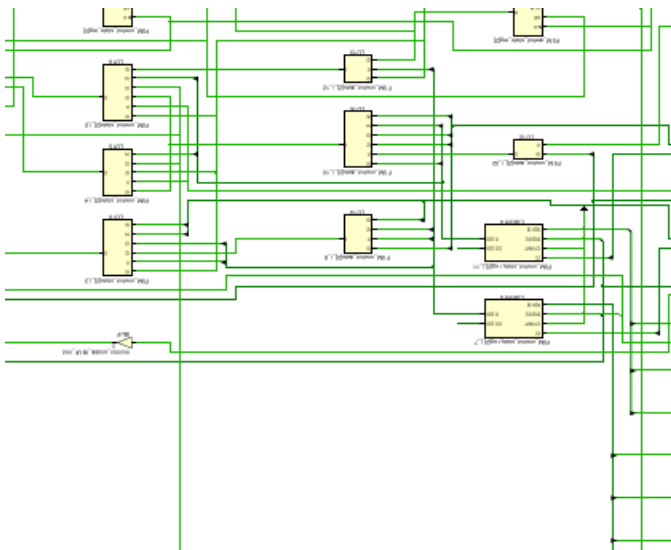


Fig. 6: Synthesized schematic – middle section.

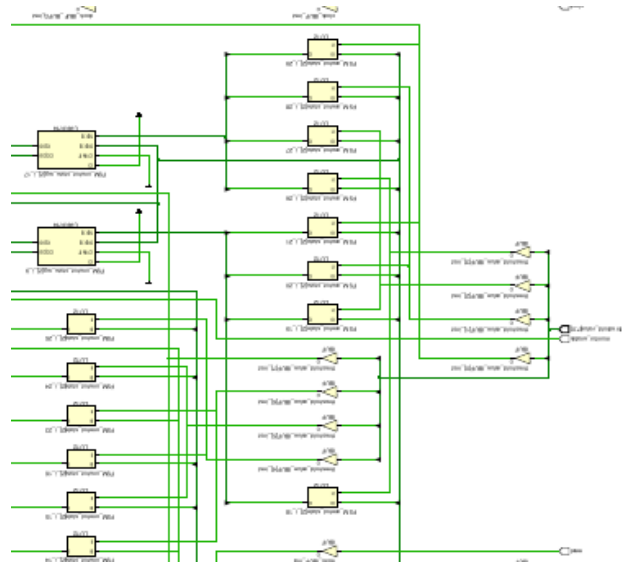


Fig. 7: Synthesized schematic – lower section.

The synthesized output shows technology-specific resources rather than generic RTL operators. The state machine is represented using flip-flop resources, while combinational decision logic is mapped into LUTs and carry-chain resources.

C. Resource Utilization

Table II summarizes the Vivado post-synthesis utilization result for the selected Spartan-7 device.

TABLE II: Post-synthesis resource utilization

Resource	Used	Available	Util.
Slice LUTs	26	14,600	0.18%
Slice Registers	12	29,200	0.04%
Bonded IOBs	30	150	20.00%
BUFGCTRL	1	32	3.13%

The register and LUT usage being very low signifies that the monitoring engine is small when compared to the Spartan-7 chosen. The utilization of I/O is greater than the mentioned figures due to the fact that module Phase-I at the top level reveals 8 bits of sensor data, 8 bits of threshold data, control inputs, and diagnostic outputs.

Name	Slice LUTs (14600)	Slice Registers (29200)	Bonded IOB (150)	BUFGCTRL (32)
data_monitor	26	12	30	1

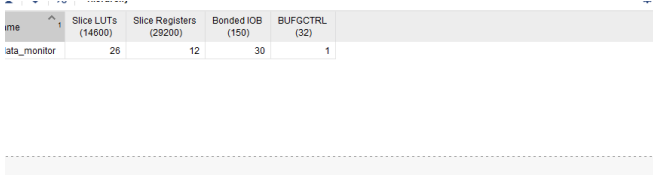
Fig. 8: Vivado post-synthesis utilization report.

D. Timing Analysis

A 10 ns clock constraint was added through the XDC file for timing analysis. The resulting Vivado timing summary is shown in Fig. 9 and summarized in Table III.

TABLE III: Vivado timing summary at 10 ns clock period

Metric	Result
Clock period	10.000 ns
Target frequency	100 MHz
Worst Negative Slack (WNS)	+6.720 ns
Total Negative Slack (TNS)	0.000 ns
Worst Hold Slack (WHS)	+0.159 ns
Total Hold Slack (THS)	0.000 ns
Failing setup endpoints	0
Failing hold endpoints	0
Worst pulse-width slack	+4.500 ns



me	Slice LUTs (14800)	Slice Registers (29200)	Bonded I/OB (150)	BUFGCTRL (32)
ata_monitor	26	12	30	1

Fig. 9: Vivado timing summary after applying the 10 ns clock constraint.

The presence of WNS and WHS values that are positive, along with the absence of failing endpoints, shows that the designed circuit is compliant with the 10 ns timing constraint. The reported results refer to the chosen target for the synthesis of Spartan-7 technology and the specified timing constraint.

VIII. APPLICATIONS AND SDG MAPPING

The Phase-I report identifies several application domains for the DEMU. Table IV retains the original mapping.

TABLE IV: Phase-I application domains and SDG mapping

Application Domain	Mapped SDG	Justification
Industrial motor overheat protection	SDG 9	Innovation in resilient infrastructure.
Automotive battery management	SDG 7	Promoting clean energy transport efficiency.
Medical fever detection / infusion pump	SDG 3	Monitoring for health and well-being.
Environmental flood warning systems	SDG 11	Safeguarding cities against climate disasters.
Agriculture soil moisture / irrigation control	SDG 6	Optimizing water-usage efficiency.
Robotics collision detection / current surge	SDG 9	Advanced manufacturing automation.
Consumer smart thermostat	SDG 12	Responsible consumption through energy saving.
Security perimeter intrusion alarm	SDG 11	Creating safe public spaces.

IX. RESULTS AND DISCUSSION

The completion of Phase-I design took place through the entire RTL to synthesis evaluation in Vivado. The behavioral simulation shows the correct functioning of the alarm system and the capturing of any fault, while RTL expansion shows what is going on within the comparison process and the operation of the FSM. The synthesis transforms the logical design into Spartan-7 features and gives quantitative information about its use and timing.

The data received after the synthesis workload is small, showing only 26 LUTs and 12 slice registers utilized in the core. Moreover, the timing experiment shows a positive slack of setup and hold complying with the applied 100 MHz constraint; thus, proving that Phase-I has successfully achieved the aim of developing a compact monitoring IP suitable for usage in larger digital systems.

The initial Phase-I report assumes that an SPI interface will be integrated along with a moving average filter for operation in noisy sensor environments. These intentions are quite reasonable after achieving validation of the standalone monitoring core.

X. CONCLUSION

The Digital Event Monitoring Unit Phase-I makes use of an 8-bit threshold-based monitoring system. It works in signed and unsigned mode, with stickiness of alarms, fault capture tracing, and a three-level FSM. The design went through eight scenarios of verification in the Verilog testbench and subsequent assessment in AMD/Xilinx Vivado 2024.1.

The output of the Vivado test adds verification information on the design in addition to Phase-I simulation results. For Spartan-7 XC7S25CSGA225-1 as a target, the design utilizes 26 LUTs, 12 slice registers, 30 bonded I/O ports, and one BUFGCTRL. With 10 ns of timing constraint, the design achieves +6.720 ns WNS and +0.159 ns WHS, with no endpoints failing. No FPGA physical board was involved in the experiment.

REFERENCES

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